PROJECT AND TEAM INFORMATION

Project Title

The Detective's Dossier

Student / Team Information

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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

Murder mysteries and detective stories have always intrigued us because of the clues,suspects,observations and the intensity of it all.Growing up watching shows like CID and always having dreamt of solving a murder mystery added to our motive and hence we decided to create this project in the form of an interactive game. In our project "The Detective's Dossier", players will take on the role of a detective in a 2D murder mystery game.They will have to collect clues,visit various locations,examine evidences and question the suspects.They will have to connect all the clues find the culprit. Our primary reason to select this project was that it allows a practical use of Data Structures and Algorithms.Several DSA concepts can be used in the investigation.

State of the Art / Current solution (1 pt)

Detective Murder Mystery games which are exiting often provide the players with the built-in cases which involves the clues, suspects,scenes and puzzles,this makes the players to find the culprits easily ,by just following the information .Thus, it fails to make the player feel like they are actually doing the detective work.
The Detective's Dossier is designed as a more structured investigation game,where the person can actually actively explore the crime scenes and collect the clues and evidences.This game involves the practical implementation of the concepts of both Data Structures and Algorithms as well as Object Oriented Programming with C++. Also ,the algorithms such as traversal algorithm(BFS,DFS),hashing,stacks,queues and graph-based structures plays an important role as they can support navigation, represent locations and relationships and in efficient investigation management.

Project Goals and Milestones (2 pts)

Main Goals:

- *To set up a 2D environment in which players can look into various murder cases.*
- *To let players visit different locations and gather evidences.*
- *To allow interactions between the player and the suspects through questioning.*
- *To include Data Structures and Algorithms as an important part of our project's logic.*
- *To organise different game entities and modules through Object Oriented Programming.*
- *To use SQLite to store all the data and player's details.*
- *To provide a result based on the player's final choice.*

Milestones:

Phase I - Planning and Design

- *Understand the problem and the requirements of the project.*
- *Finalise the game's concept and features.*
- *Choose the technologies and various concepts that are to be used.*
- *Draw the architecture of game and design the project.*

Phase II - Development

- *Create a 2D interface using Qt.*
- *Build the OOP structure of the game and the logic using DSA.*
- *Set up SQLite database and work on the key aspects of investigation and gameplay.*

Phase III - Integration and Testing

- *Make sure that all modules are connected.*
- *Add some murder cases and test them.*
- *Check DSA implementation and investigate the entire game.*
- *Conduct user testing and make improvements to the final prototype.*

Project Approach (3 pts)

The system is build as a 2D interactive investigation game with the application of Object Oriented Programming with C++ ,Qt and SQlite.The project follows a modern approach in which it involves Graphical User Interface,Data Structures and Algorithms and application framework. • ***User Interaction***

The player will start the game by a login interface. After the successful login,the player will be able to the access the available investigation cases and continuing investigation process.

- ***2D Game Interface***

Qt will be used for the development of the investigation game. The interface will give the case selections , locations to explore, evidence to inspect , find the suspects, manage the clues.

- ***C++ and OOP***

The application will be implemented with the concepts of Object-Oriented Programming with C++ . use of appropriate encapsulation, inheritance and abstraction for the design

- ***DSA-Driven Investigation*** *Graph:*

The game will represent the relationship relationships between various evidences, suspects, locations through graphs. For example, if a suspect is connected to a location then that location can contain some evidences, and those evidences can lead us to new suspects. This will allow the player to build and explore connections amongst various parts of the game.

DFS:

DFS will be used when a player follows a connection of clues .when it starts from one clue or suspect , DFS will explore the connected entities and help to find the information.

BFS:

BFS will be used when the game needs to explore the connected components level by level and it would be useful when a player wants to find nearby connected information.

Hashing:

Since all the suspects, clues and evidences will have unique IDs. therefore,, a hash table can be used to store all these details and quickly find the required information whenever the player selects any suspect, clue or evidence.

Stack:

Stack will keep a record of the players, investigation cases . Whenever the player performs any action, it will be added into the stack. Also allow the players to return to their previous investigation details.

Searching:

Searching techniques will be used when a player wants to look for a particular suspect, clue, evidence , or location. and depending on how the data is stored , linear or binary search can be used.

Sorting:

Sorting will be used to arrange suspects or evidences according to properties like suspicion level, importance, or time found. This can help the player analyse the collected information more easily.

- ***Database and Storage***

Sqlite is the local database being usd for the storage for the user account, cases, clues and locations.

- ***Development and Testing***

The project will be carried out step by step by first creating and testing each module separately before integrating them.

System Architecture (High Level Diagram)(2 pts)

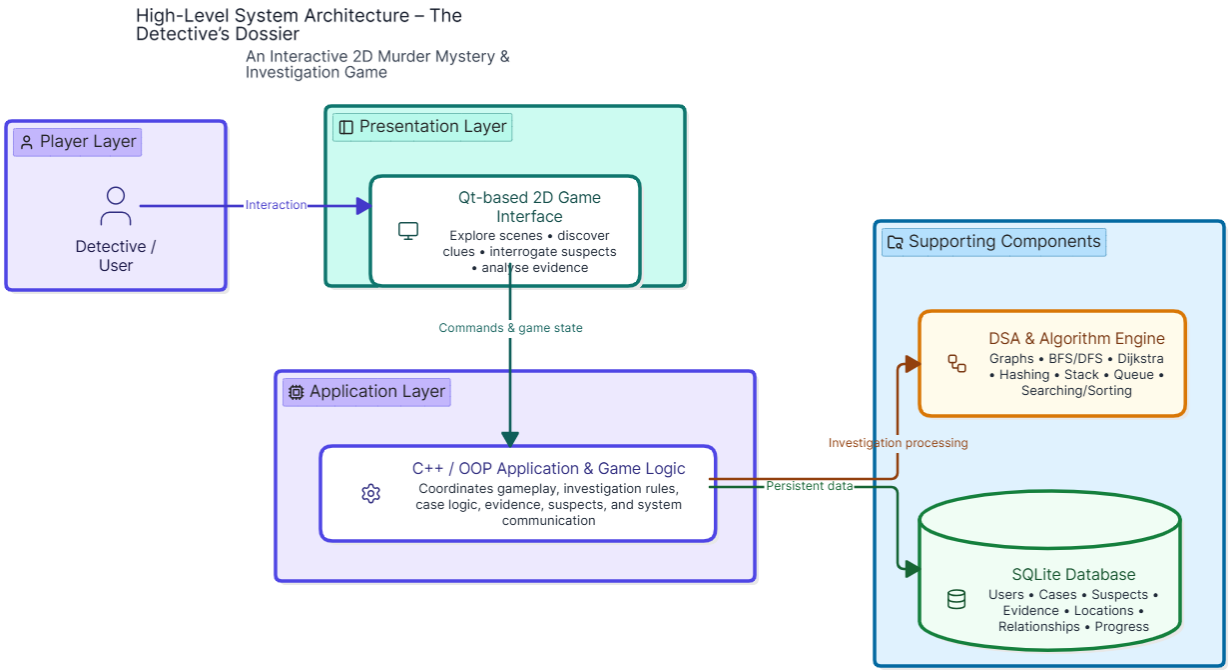


Figure 1: High-Level System Architecture of The Detective's Dossier

Project Outcome / Deliverables (1 pts)

Expected Deliverables

- *A working 2D user interface using Qt.*
- *A game structured using OOP concepts.*
- *An investigation and navigation system based on DSA.*
- *A number of murder mysteries that can be solved by the player.*
- *Interaction between player,suspects,evidences and locations.*
- *A system for final accusation and outcome of the case.*
- *A working prototype that has been tested and integrated.*

Assumptions

- *It will be a desktop based 2D interactive investigation game.*
- *It will run on sqlite and qt.*
- *The local database will store the clues, evidences and locations.*
- *It will select the dsa implementation according to the requirement of the gameplay and investigation .*

References

1. *C++ Reference, cppreference:* `https://en.cppreference.com/`
2. *Qt Documentation:* `https://doc.qt.io/`
3. *SQLite Documentation:* `https://www.sqlite.org/docs.html`
4. *T. H. Cormen, C. E. Leiserson, R. L. Rivest and C. Stein, Introduction to Algorithms, MIT Press.*
5. *R. Lafore, Object-Oriented Programming in C++.*
6. *Qt Creator Documentation:* `https://doc.qt.io/qtcreator/`